



Intro to the web

Web programming I



Intro to the web

Student is able to

identify and explain the principles of client–server architecture and the HTTP protocol.



Content

Internet ja web

What is the difference between the Internet and the Web?

How did the Web evolve over time?

Client-server architecture

What is the client-server architecture?

HTTP

What is HTTP, and how does it work?

What HTTP methods and response codes do we need to know?



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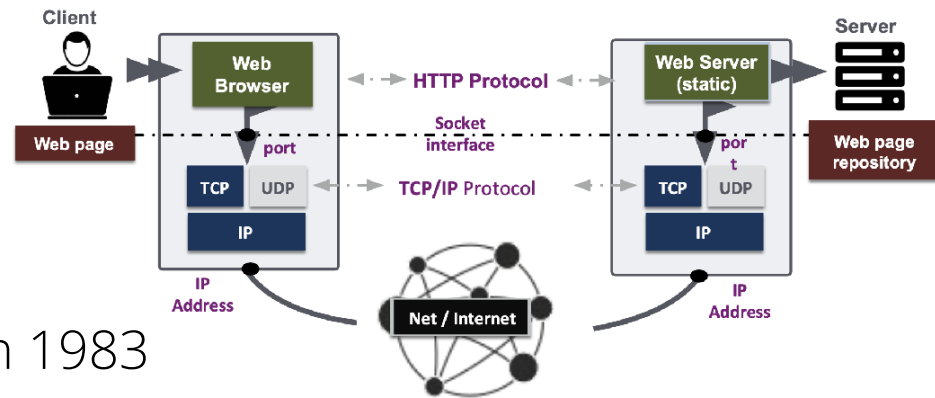
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Origin of the Internet

- The Internet started as ARPANET in 1983
- The Internet connects computers to the network using the TCP/IP protocol
- TCP/IP supports client-server applications through a socket interface
 - The client-server model is a distributed application structure that divides the tasks to be performed between resource or service providers, called servers, and those interested in these services or products, called clients.
- The first applications on the Internet were: *telnet* (virtual terminal), *ftp* (file transfer), *email*, ...





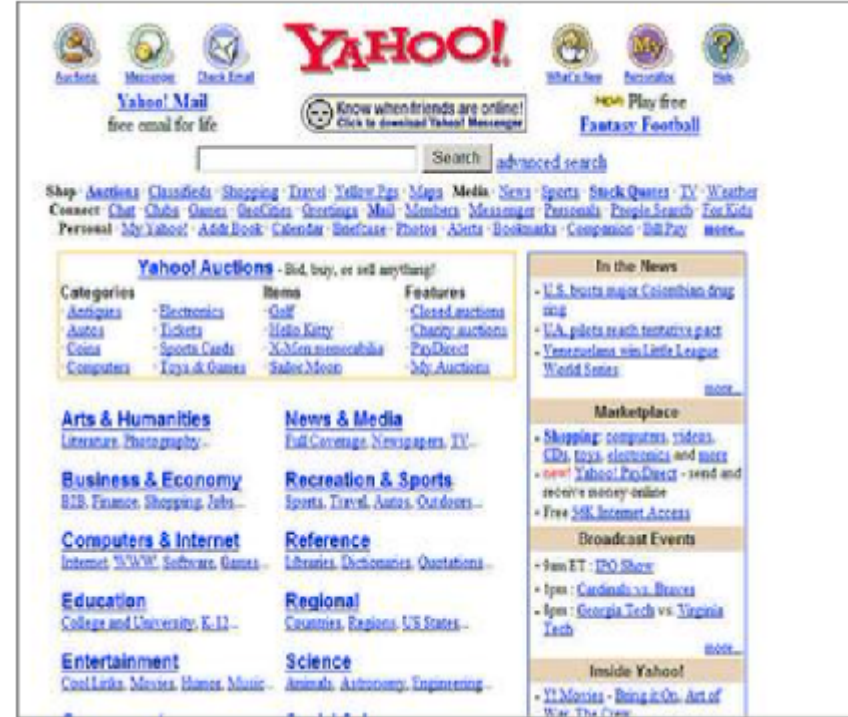
WWW – the Web

- Tim Berners Lee proposed a new application in 1989: the World Wide Web
- The Web is an information space where we find hypertext documents and other resources uniquely identified by their Uniform Resource Locator (URL), interconnected by hyperlinks and accessible through the Internet.
- There is a client application (text browser) and a server application (static web server)
- Tim Berners Lee developed/joined three essential technologies:
 - URL: Uniform Resource Locator – address of a resource
 - HTML: Hypertext Markup Language – structure of content
 - HTTP: Hypertext Transfer Protocol – communication protocol



The Web in the 90s

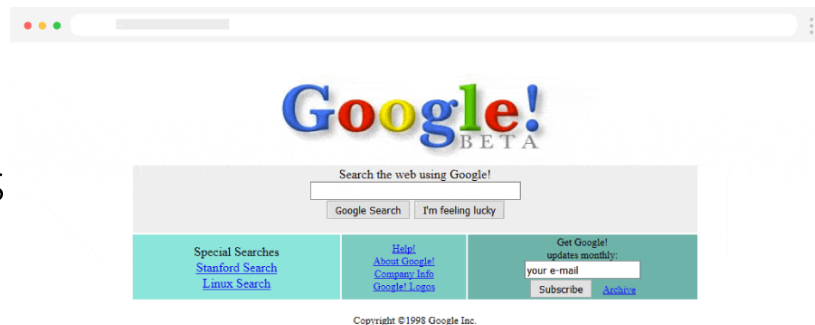
- **Static** pages are served (HTTP GET) and forms are sent (HTTP POST).
- With each request, all displayed content is **refreshed**: complete pages.
- Logic is only on the **server side**.





Dynamic Web in late 1990s

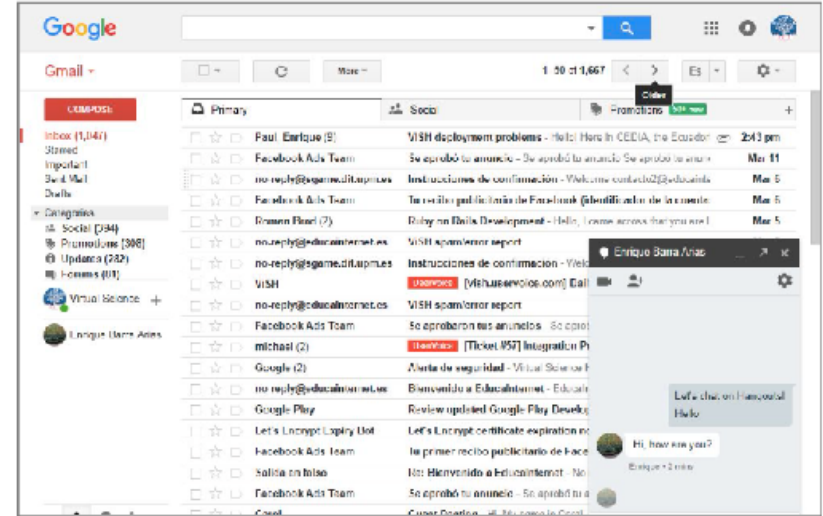
- Pages were **generated** on the server per request
- Often using PHP or ASP, with cookies and databases enabling **user-specific customization**.
- Each interaction still caused a **full-page reload**, but what appeared was increasingly personalized and responsive.





The Web in the 2000s

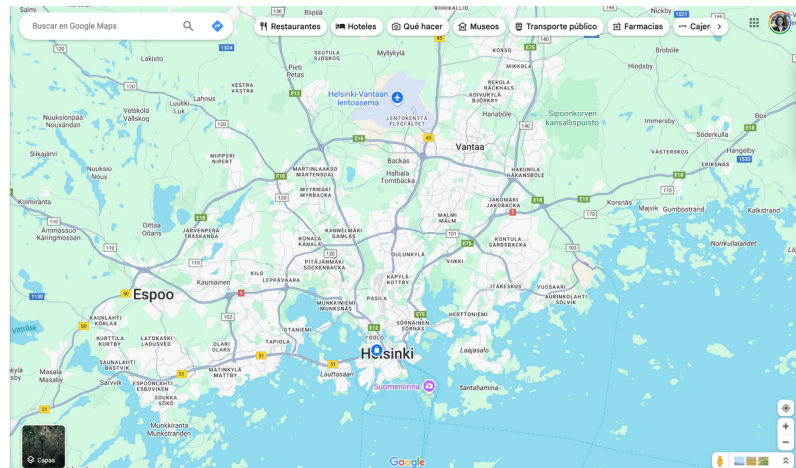
- Fetching **pieces** of pages without reloading the entire page.
- **AJAX** (Asynchronous JavaScript and XML) - XMLHttpRequest
- Use of **JavaScript** to communicate asynchronously with the server.
- Most of the logic is on the server.





Today's Web

- **SPA** (Single Page Application)
- Web application architecture. Similar to desktop applications.
- Most of the logic is on the **client**.
- The server is only for storing data and authentication and not much else.
- With the first request the browser fetches the web application, and later other technologies are used to communicate with the server and send and receive data from the server.
 - AJAX, Websockets, RTSP, WebRTC, ...





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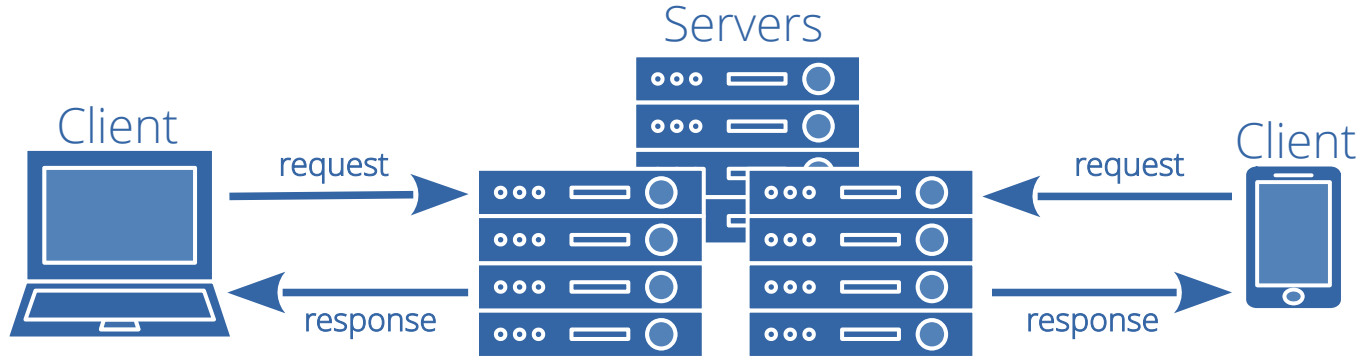
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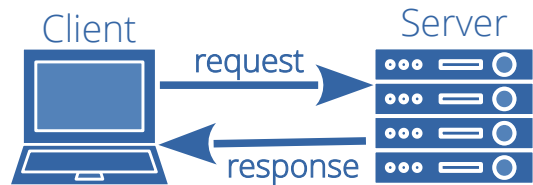
Overview



The Client makes request for service to server.
The service responds to the request.



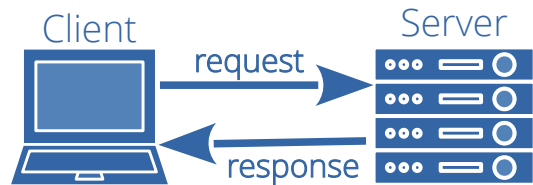
Servers



- **machine** that serves the content
 - a physical machine or a virtual machine in the cloud
 - server machine has a “known” address on the Internet (DNS name resolution-> IP)
- **program** that provides services to clients is also called a server.
 - Server programs run on a server machine.
 - Each program connects to a port on the server machine (the Web server uses port 80 by default).
 - Examples of web servers: Apache, Nginx, Microsoft-IIS, Nodejs, etc.



Clients

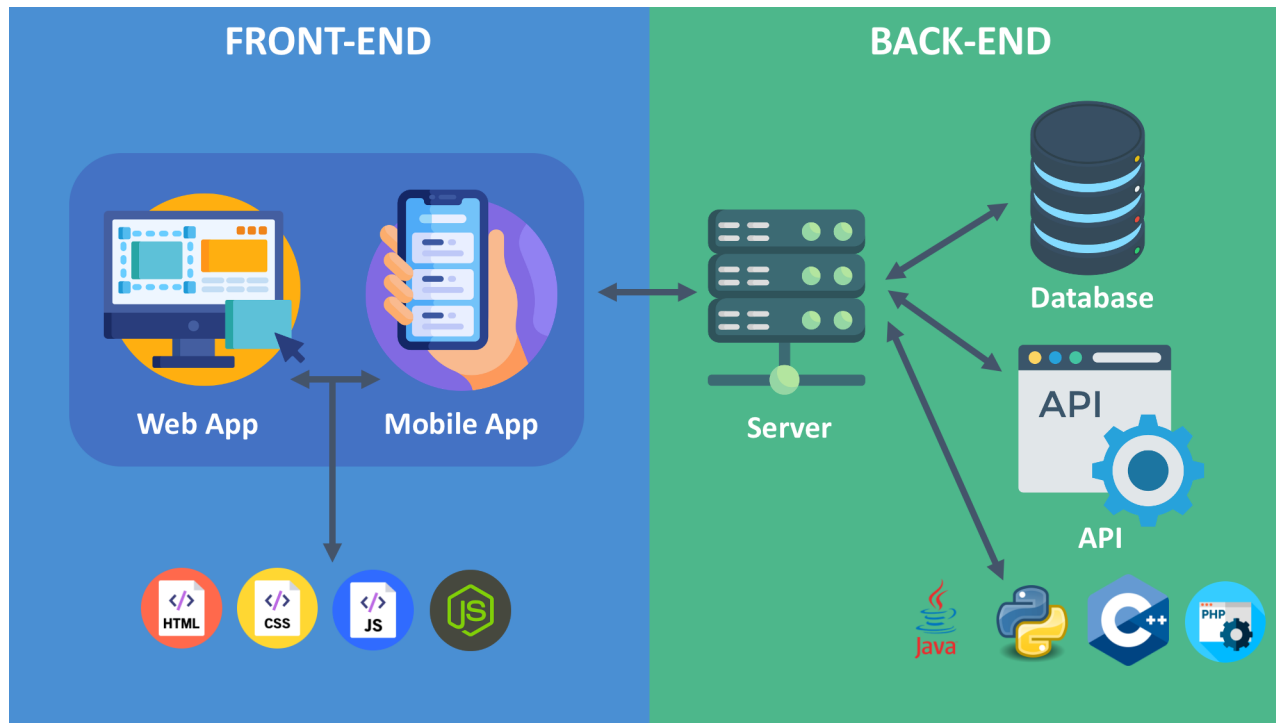


- **devices** that connect to the Internet: PCs, laptops, tablets, phones, etc.
- **programs** that access services on the Internet
 - **browser** is the main access client: Chrome, Firefox, Opera, Safari, ...
 - content served on the browser is programmed in HTML, CSS and JavaScript.
 - **mobile** device apps are the most used clients today.
 - Native applications (apps): Android, iOS, etc.
 - programmed in development environments with specific languages: Android programmed in Java/Kotlin, iOS with Swift, etc.



Client-server architecture > Client-server on web development

Client-server on web development





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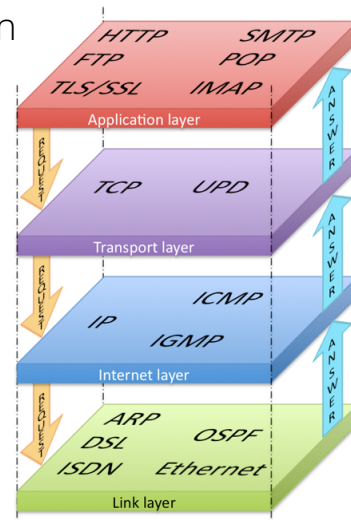
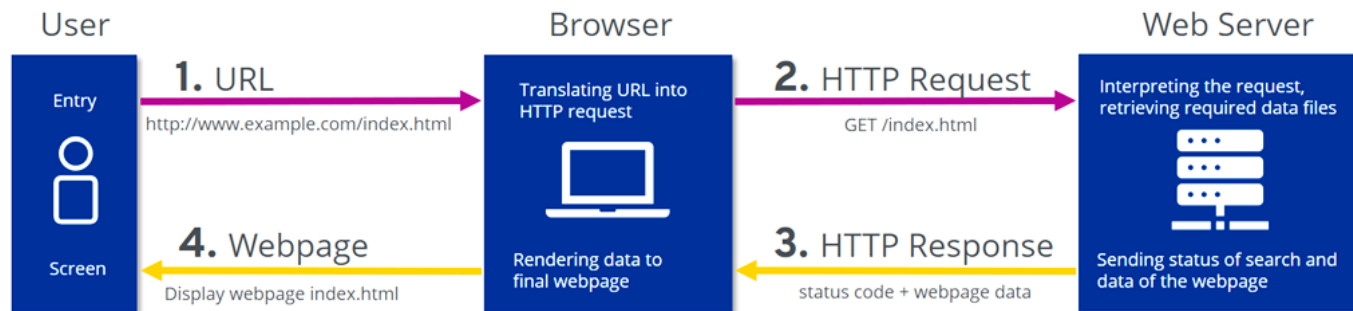
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Hypertext Transfer Protocol

- supports hypermedia access to Web pages and other Internet services
- stateless transactional request-response protocol
 - HTTP transaction is always initiated from the client side
- the client establishes HTTP connection with the server or reuses an existing one
- the client sends HTTP Requests to the server over the HTTP connection
- the server answers the client by sending HTTP Responses over the connection
 - request and response are independent, though related, transmissions
 - the request and response have the same format





HTTP methods

The type of primitive of an HTTP request is indicated by the method or verbs:

- Uniform interface, methods allow you to manage resources on the server with the four CRUD primitives:
 - **POST**: Create a resource on the server.
 - **GET**: Read a resource from the server.
 - **PUT**: Update a resource from the server.
 - **DELETE**: Delete a resource from the server.



HTTP Responses

Informative Responses (1xx)

- 100 Continue: Continue partial request

Request completed (2xx)

- 200 OK: Operation completed successfully, resource served
- 201 Created: Resource created or updated successfully

Redirect (3xx)

- 301 Moved Permanently: Resource on another server
- 304 Not Modified: When the client already has the data

Client Error (4xx)

- 400 Bad Request: Incorrect command sent
- 404 Not Found: Resource not found in repository
- 405 Method Not Allowed: Method not supported by server
- 409 Conflict: Conflict with the resource status
- 410 Gone: Resource no longer exists

Server Error (5xx)

- 500 Internal Server Error: The server has errors



MIME type (Multipurpose Internet Mail Extensions)

- defines the type of a resource or file in email, Web,
- format: "type/subtype":
https://developer.mozilla.org/ar/docs/Web/HTTP/Basics_of_HTTP/MIME_Types
- most common MIME types for the Web

– Text type:

<code>text/plain</code>	Plain text without format
<code>text/html</code>	HTML
<code>text/css</code>	CSS file
<code>text/javascript</code>	JavaScript script

– Image type:

<code>image/gif</code>	Image in GIF format
<code>image/jpeg</code>	Image in JPEG format
<code>image/png</code>	Image in PNG format
<code>image/svg</code>	Image in SVG vector format

– Sending form parameters in the body:

<code>application/x-www-form-urlencoded</code>	Parameters encoded in URL-encoded
<code>multipart/form-data</code>	Parameters with multiple formats

– Server response with multiple blocks and parts:

<code>multipart/byte-ranges</code>	Response in multiple blocks
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Or use QR code



Summary

- The Internet is the underlying network infrastructure, the global network of computers connected through TCP/IP; the Web is one application built on top of the Internet
- The Web has evolved from static pages, to dynamic server-rendered content, to Single Page Applications where most logic runs on the client.
- The client-server architecture divides tasks: clients (browsers, mobile apps) request services; servers provide content, data, and business logic.
- HTTP is a stateless, request-response protocol. The client always initiates the request, and the server responds.
- HTTP uses standard methods (GET, POST, PUT, DELETE) to manage resources, and response codes (1xx, 2xx, 3xx, 4xx, 5xx) to indicate the result.



Bibliography

- Materials of the course *Computación en Red* - Universidad Politécnica de Madrid